Part 2 Brief · Core server

mcp-ops-server · 18 Oct 2026 · PR #3 · Status: CI green on push and PR (56 tests), awaiting Codex review

WHAT WAS BUILT

- **Synthetic clinic data** (`seed.py`, fixed seed): 12 providers across 6 specialties (4 need a referral), 40 patients, 2,688 thirty-minute slots over 4 weeks, 42 referrals, 128 pre-booked appointments. The same seed always gives the same IDs.
- **Five tools** over a SQLite store: `search_providers`, `list_open_slots`, `book_appointment`, `cancel_appointment`, `get_referral_status`.
- **56 tests** (up from 3), including a 20-thread race for one slot and a real HTTP client running the full find, check, book, cancel flow.

HOW IT WORKS

Layer	Role
tools/	Thin wrappers. Typed inputs (enums, ID patterns, bounded limits) and a TypedDict output for each tool. Store errors become ToolErrors whose message names the next tool to call.
store.py	All business rules. Booking checks: slot open, patient free at that time, approved referral owned by this patient and matching the specialty. One check-then-insert transaction.
SQLite schema	Partial unique index `(slot_id) WHERE status='booked'`: the database itself refuses a second booking, so a cancelled slot can be rebooked but an open one is never booked twice.

WHY IT WAS DONE THIS WAY

- **Rules in the store, constraint in the schema.** The app check gives agents a clear error; the index makes double booking impossible even if a future code path forgets the check.
- **Cancelling needs the owner's patient ID**, and someone else's appointment gives the same "not found" as a fake ID, so the tool can't be used to probe which appointments exist.
- **Deterministic, clock-free data**, so Part 4 eval tasks can name exact slots and referrals. Trade-off: booking past slots is not blocked yet.

WHAT WENT WRONG (LOGGED IN FAILURES.MD)

- **Threads.** The SDK runs tools in worker threads. A mutation test with the store lock removed crashed with `sqlite3.InterfaceError`. Kept the lock and added BEGIN IMMEDIATE.
- **Regex.** pydantic's `\d` matches any Unicode digit, so a patient ID written in Arabic-Indic digits passed validation. Switched to `[0-9]` and added a test.
- **Fixtures.** 5 tests first failed because the test patient already had a seeded booking at that time. Now the tests use patients with clean calendars, pinned by a seed test.

VERIFY IT YOURSELF

```
pip install -e ".[dev]"
ruff check . && python -m pytest -q   # 56 passed
python -m mcp_ops_server              # then call tools at http://127.0.0.1:8000/mcp
```

INTERVIEW TALKING POINT

"Double booking is prevented twice: once in application code so the agent gets an actionable error, and once as a partial unique index so the database refuses it even under a race. I proved the race test catches a broken implementation by deleting the protection and watching it fail."

NEXT: PART 3, HARDENING

Bearer auth and scopes enforced in the server, rate limits, payload size and timeouts, and a "schema drift" snapshot test that flags any change to a tool schema. **Your action:** paste the Codex prompt, merge PR #3 once review fixes are in, then reply "go".